Shan Yang

Staff Applied Scientist — Foundation Models, Multimodal AI, Generative AI

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Professional Summary

- Technical leader with 8+ years driving foundation model research and production ML systems at scale across multimodal generative AI, video/image generation, and LLM post-training
- Proven track record managing Applied Scientists and establishing technical roadmaps, shipping products serving millions with $500M+ revenue impact
- Deep expertise in multimodal foundation models: 3,500+ citations at premier venues (NeurIPS, ICCV, ECCV, ICML, AAAI), pioneering novel architectures in attention-based fusion, music-conditioned motion synthesis, and autoregressive diffusion

Technical Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-learn — Distributed training (DeepSpeed, FSDP)

Foundation Models: Diffusion Models, Multimodal LLMs, Transformers, CLIP, BLIP

Research Focus: Novel Model Architectures, Image/Video/3D Generation & Editing, Multimodal Understanding, Post-Training (RLHF), Test-Time Training, Reinforcement Learning

Architecture Innovation: Attention Bottlenecks, Multimodal Fusion, Music-Conditioned Motion Synthesis, Autoregressive Diffusion, Linear Recurrent Models

Engineering & Research Experience

Adobe Foundry — San Jose, CA

Staff Applied Scientist — March 2026 – Present

- Driving post-training research for Adobe Foundry, Adobe's family of generative AI foundation models powering Firefly and creative tooling at scale
- Developing post-training techniques (SFT, RLHF/DPO, LoRA, distillation, test-time training) to align foundation models with creator intent and production quality bars

Amazon (Prime Video & Search) — Palo Alto & Sunnyvale, CA

Senior Applied Scientist Tech Lead — September 2021 – March 2026

- **Prime Video (2025–2026):** Led GenAI Live Action Studio with **5 Applied Scientists**, building autoregressive diffusion architectures and training infrastructure for 10B+ parameter video generation models; advanced controllable generation via novel post-training (Test-Time Training, Teacher-Diffusion-Forcing, RL)
- **Search (2021–2025):** Led **Amazon Video Search** from zero to launch serving millions with **$500M+ annual revenue gain**, managing **3 Applied Scientists and 2 MLEs** and influencing a 15-person SDE team; pioneered multimodal fusion architecture integrating video, text, image, and structured data
- Drove multimodal LLM robustness research: Schema-of-Thought framework achieving **+28% hallucination reduction**; published at ECCV, AAAI and filed multiple patents

Google Research — Mountain View, CA

Senior Research Software Engineer — June 2017 – September 2021

- Led foundational research in multimodal generative models, influencing **3 Research Engineers**, driving research agenda in music-conditioned 3D motion synthesis and video-audio-motion modeling, publishing at premier ML conferences (ICCV, NeurIPS, ECCV)
- Drove technical ownership of 2D/3D pose reconstruction models deployed on GCP at scale, demonstrating ability to lead end-to-end ML product development from research to production
- Provided technical leadership through patent contributions, cross-functional collaboration, and mentorship of research interns on advanced ML projects

Research in Progress

Multimodal Reasoning RL: Reinforcement-learning post-training of vision-language reasoning models; physics olympiad benchmark and reference recipe published as Physics-R1 (arXiv 2026).

Selected Publications

Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning
S Yang
arXiv 2026 (2605.14040)
Sole-author preprint. Three-stage corpus audit (Jaccard → embedding → LLM-judge), novel-source held-out olympiad benchmark (PhysOlym-A, 500q), and GSPO+DAPO recipe lifting Qwen3-VL-8B by +18.3 pp on PhysOlym-A.

Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control
*P Wang, **S Yang**, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal*
ICML 2026
Linear recurrent architecture leveraging optimal control theory for LLM test-time reasoning (+35.8% Math-500, +10% AIME25).

Schema Perception for Robust Video Question Answering
*X Wang, **S Yang***
Internal 2025
Schema-of-Thought prompting framework for multimodal LLM robustness and OOD generalization (+28% Claude 3.5, +13.3% GPT-4o).

ViLA: Efficient Video-Language Alignment for Video Question Answering
*X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, **S Yang***
ECCV 2024
Efficient vision-language multimodal foundation models.

ICAR: Image-based Complementary Auto Reasoning
*X Wang, A Liang, J Liang, M Lin, Y Lou, **S Yang***
AAAI 2024
Multimodal reasoning for visual understanding tasks.

Attention Bottlenecks for Multimodal Fusion
*A Nagrani, **S Yang**, A Arnab, A Jansen, C Schmid, C Sun*
NeurIPS 2021 — **800+ citations**
Multimodal fusion architecture for foundation models.

AI Choreographer: Music Conditioned 3D Dance Generation with AIST++
R Li, **S Yang***, DA Ross, A Kanazawa*
ICCV 2021 — **700+ citations**
Multimodal generative modeling: music, motion, vision.

Physics-Inspired Garment Recovery from a Single-View Image
***S Yang**, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin*
ACM ToG 2018 — **100+ citations**
3D generation from images.

Modeling Context in Referring Expressions
*L Yu, P Poirson, **S Yang**, A Berg, T Berg*
ECCV 2016 — **1600+ citations**
Vision-language understanding.

Patents & Intellectual Property

VLAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling (2023)
Monitoring Animal Pose Dynamics from Monocular Images (2022)
Image-based Complementary Item Recommendation Algorithm (2022)
Auto Design: ML-based Furniture Layout Recommendation Algorithm (2022)

Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
Ph.D., Computer Science — 2012 – 2017

Shanghai Jiao Tong University — Shanghai, China
B.E., Software Engineering — 2008 – 2012